CPSC 346 — hw5

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a)

the number of virtual pages is the size of the virtual address space divided by the page size:

$$\#\text{virtual pages} = \frac{2^{\text{virtual bits}}}{\text{page size}} = \frac{2^{\text{virtual bits}}}{2^{\text{offset bits}}} = 2^{\text{virtual bits}-\text{offset bits}}$$

the offset needs $\log_2(\text{page size})$ bits, and the page number uses the remaining bits. so the maximum number of virtual pages is 2^p

	design 1	design 2	design 3
virtual address bits	16	32	36
page size	$32 = 2^5$	$64 = 2^6$	$128 = 2^7$
offset bits $\lvert d\rvert$	5	6	7
page number bits $\lvert p\rvert$	$16 - 5 = 11$	$32 - 6 = 26$	$36 - 7 = 29$
max virtual pages	$\mathbf{2^{11}}$	$\mathbf{2^{26}}$	$\mathbf{2^{29}}$

- **design 1:** $\frac{2^{16}}{2^5} = 2^{11} = \mathbf{2{,}048}$ pages
- **design 2:** $\frac{2^{32}}{2^6} = 2^{26} = \mathbf{67{,}108{,}864}$ pages
- **design 3:** $\frac{2^{36}}{2^7} = 2^{29} = \mathbf{536{,}870{,}912}$ pages

b)

the number of physical pages is the size of physical memory divided by the page size:

$$\#\text{physical pages} = \frac{2^{\text{physical bits}}}{\text{page size}} = 2^{\text{physical bits}-\text{offset bits}}$$

- **design 1:** $\frac{2^{10}}{2^5} = 2^5 = \mathbf{32}$ pages
- **design 2:** $\frac{2^{20}}{2^6} = 2^{14} = \mathbf{16{,}384}$ pages
- **design 3:** $\frac{2^{24}}{2^7} = 2^{17} = \mathbf{131{,}072}$ pages

c)

splits into outer, inner, and offset:

- **outer:** 128 entries = 2^7, so **7** bits
- **offset:** 128-byte page = 2^7, so **7** bits
- **inner:** the rest, $36 - 7 - 7 = \mathbf{22}$ bits